

■ 3.6.7 Finding Limits

In doing many kinds of calculations, you need to evaluate expressions when variables take on particular values. In many cases, you can do this simply by applying transformation rules for the variables using the /. operator.

You can get the value of $\cos(x^2)$ just by explicitly replacing x with 0, and then evaluating the result.

```
In[1]:= Cos[x^2] /. x -> 0
Out[1]= 1
```

In some cases, however, you have to be more careful.

Consider, for example, finding the value of the expression $\frac{\sin(x)}{x}$ when $x = 0$. If you simply replace x by 0 in this expression, you get the indeterminate result $\frac{0}{0}$. To find the correct value of $\frac{\sin(x)}{x}$ when $x = 0$, you need to take the *limit*, in this case using L'Hospital's rule.

`Limit[expr, x -> x0]` find the limit of *expr* when *x* approaches *x0*

Finding limits.

This gives the correct value for the limit of $\frac{\sin(x)}{x}$ as $x \rightarrow 0$.

```
In[2]:= Limit[ Sin[x]/x, x -> 0 ]
Out[2]= 1
```

No finite limit exists in this case.

```
In[3]:= Limit[ Sin[x]/x^2, x -> 0 ]
Out[3]= ComplexInfinity
```

`Limit` can find this limit, even though you cannot get an ordinary power series for $x \log(x)$ at $x = 0$.

```
In[4]:= Limit[ x Log[x], x -> 0 ]
Out[4]= 0
```

$\exp(\frac{1}{x})$ has an essential singularity at $x = 0$, so no limit exists.

```
In[5]:= Limit[ Exp[1/x], x -> 0 ]
Limit::nlm: Failed to compute the limit.

      1
      -
      x
Out[5]= Limit[E^ , x -> 0]
```

If you give `Limit` a function it does not know, it assumes that the function is bounded at the limit point.

```
In[6]:= Limit[ f[x], x -> 0 ]
Out[6]= f[0]
```